

PROJECT OVERVIEW: ID-Link Tanzania

Concept: A secure, digital clearinghouse for lost and found identification documents, physically anchored at the **Tanzania Post Office, Countrywide**.

PART 1: BUSINESS PROPOSAL

1.1 Problem Statement

Every month, hundreds of IDs (NIDA, Voter cards, Licenses) are lost in Tanzania. Currently, there is no central database. Owners wander from police stations to bus terminals, while finders have no official, rewarded way to return them.

1.2 The Solution

A mobile-responsive web portal where:

1. **Finders** (or Postal Staff) digitize found IDs.
2. **Owners** search, verify their identity, and pay a convenience fee.
3. **The Post Office** acts as a secure "Safe Zone" for physical collection.

1.3 Monetization Strategy

- **Recovery Fee:** 10,000 TZS (Required to generate the "Collection Token").
 - **Premium Delivery:** 15,000 TZS (Courier delivery via Post Office logistics).
 - **Subscription Alerts:** 5,000 TZS (One-time fee to get notified if an ID matching your number is found in the future).
 - **Revenue Share:** 20% to the Tanzania Post Corporation; 80% to ID-Link for maintenance and profit.
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PART 2: TECHNICAL REQUIREMENTS SPECIFICATION (TRS)

2.1 System Architecture

- **Platform:** Mobile-first Web Application (accessible via Chrome/Safari).
- **Language/Framework:** React.js or Flutter (Frontend) + Node.js or Python/Django (Backend).
- **Database:** PostgreSQL (for robust searching and data integrity).
- **Hosting:** Local servers (e.g., Arusha-based or TZ-based cloud) to comply with data residency laws.

2.2 Core Modules & Features

A. The Public Search Engine (Owner View)

- **Fuzzy Search:** Users search by "Name" or "Region Found."
- **Data Masking:** Search results must mask sensitive data.
 - *Display:* Juma H. | NIDA Card | Found: Arusha | ID: 1990*****001
- **Claim Logic:** To "Unlock" the location/contact, the user must enter the **Full ID Number**. If it matches the database, they proceed to payment.

B. The Finder/Admin Portal (Post Office View)

- **OCR Integration:** Use **Google Vision AI** or **AWS Rekognition**. When the clerk snaps a photo of the ID, the AI automatically extracts:
 - Full Name
 - ID Number
 - Date of Birth (for internal verification only)
- **Status Management:** Toggle ID status between Logged, Verified, Collected, or Transferred.

C. Payment Integration

- **Gateway:** Integration with a local aggregator (e.g., **Selcom, Pesapal, or AzamPay**).
- **Workflow:** User hits "Pay" -> Prompted for M-Pesa/Tigo Pesa PIN -> Upon success, the **Collection QR Code** is instantly displayed and emailed.

2.3 Data Security & Compliance

- **Encryption:** AES-256 encryption for ID numbers stored in the database.
- **Data Retention:** Automatic deletion of records 6 months after the "Collected" status to minimize data liability.
- **PDPA Compliance:** Explicit "Consent to Process Data" checkbox on every form.

PART 3: IMPLEMENTATION ROADMAP

PHASE	MILESTONE	DURATION
Phase 1	Legal & Partnerships: Finalize MoU with Tanzania Post Office; register with PDPC.	4 Weeks
Phase 2	MVP Development: Build the database, OCR scanning, and Search UI.	8 Weeks
Phase 3	Beta Testing: Internal testing at the Post Office counter using 50 "dummy" IDs.	2 Weeks
Phase 4	Public Launch: Social media campaign and physical signage at prime location.	Day 1

PART 4: SYSTEM DATA SCHEMA (High-Level)

DEVELOPER – **Gideon Msambwa**

PART 5: NEXT STEPS

1. **Legal:** Apply for the Data Controller certificate from the PDPC.
2. **Financial:** Open a business mobile money account (Lipa Namba) to receive fees.
3. **Technical:** Hire a developer to build the "Alpha" version based on the Part 2 specifications.